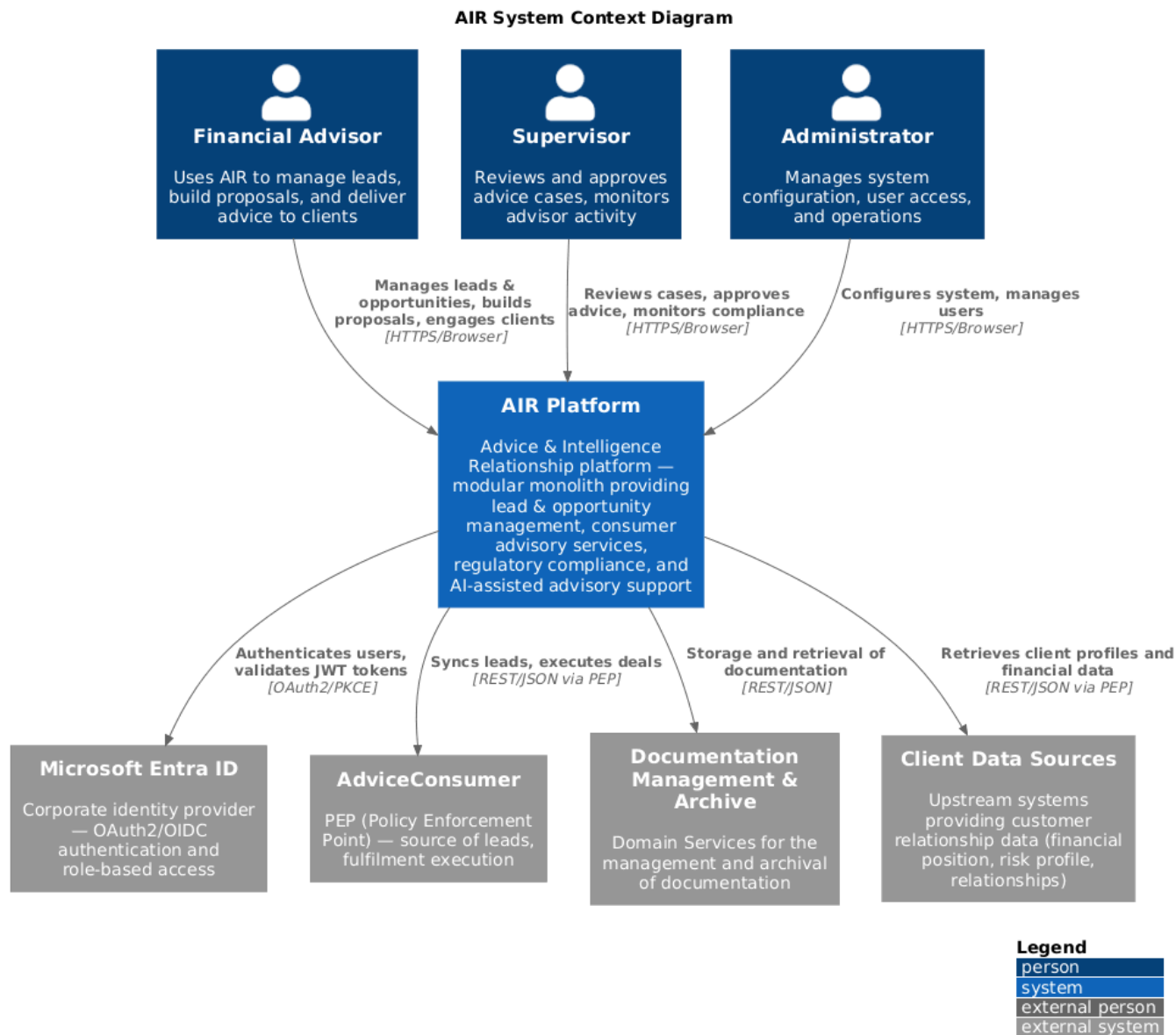


AIR (Advice & Intelligence Relationship) System Architecture Design – V2.0

Version: 2.0
Based on: System Architecture Design V1.0 + AIR DDD v3
Date: May 2026

Level 1: System Context Diagram

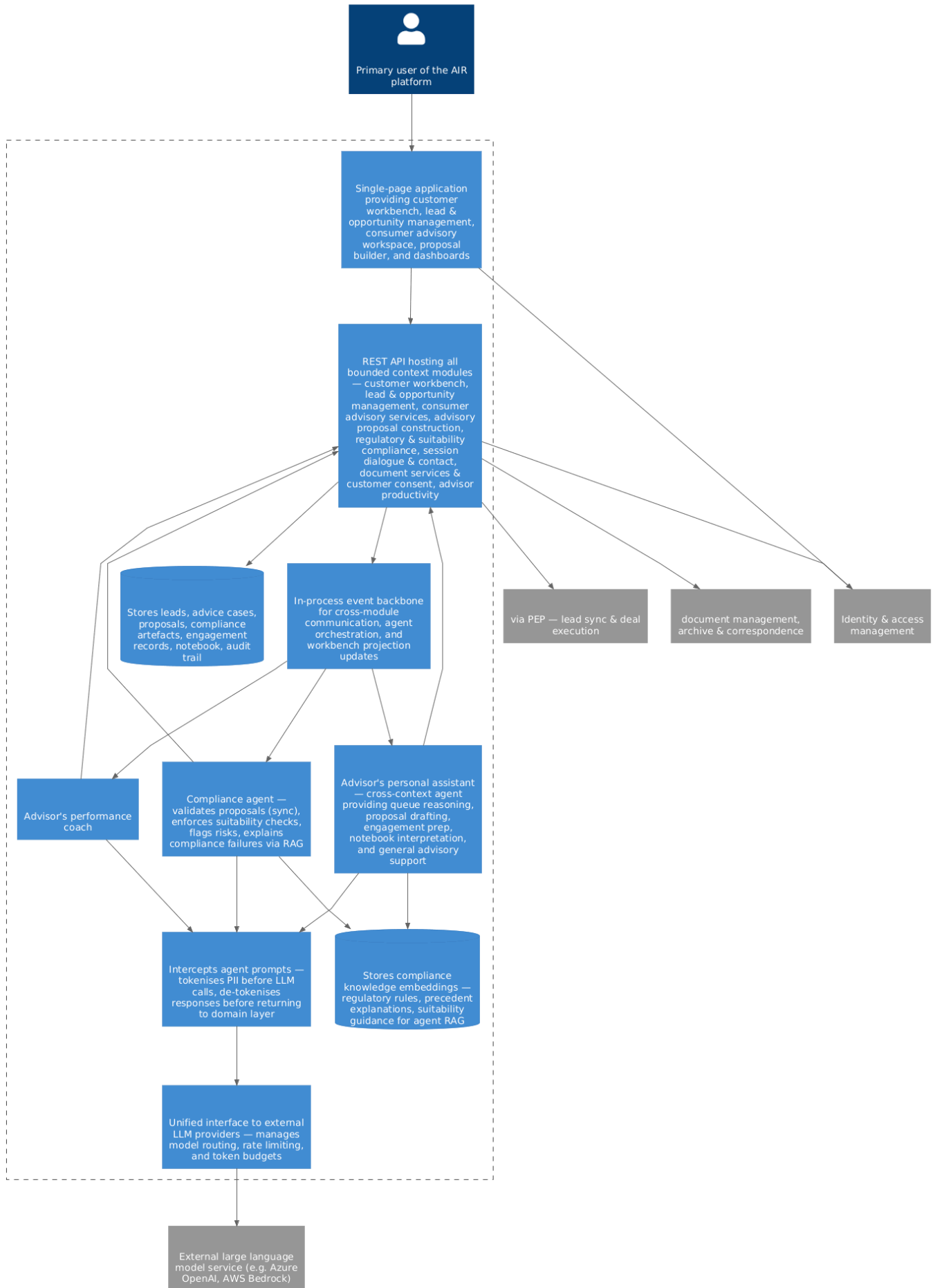
Shows the AIR system in its environment, with external actors and systems.



Level 2: Container Diagram

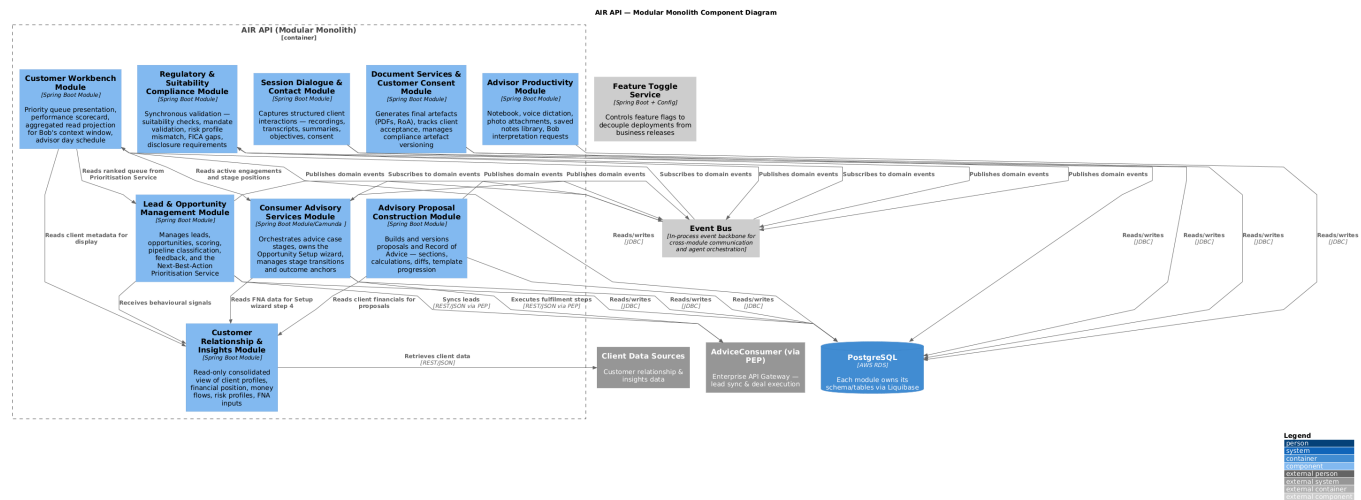
Shows the major deployable units and their interactions.

AIR Container Diagram



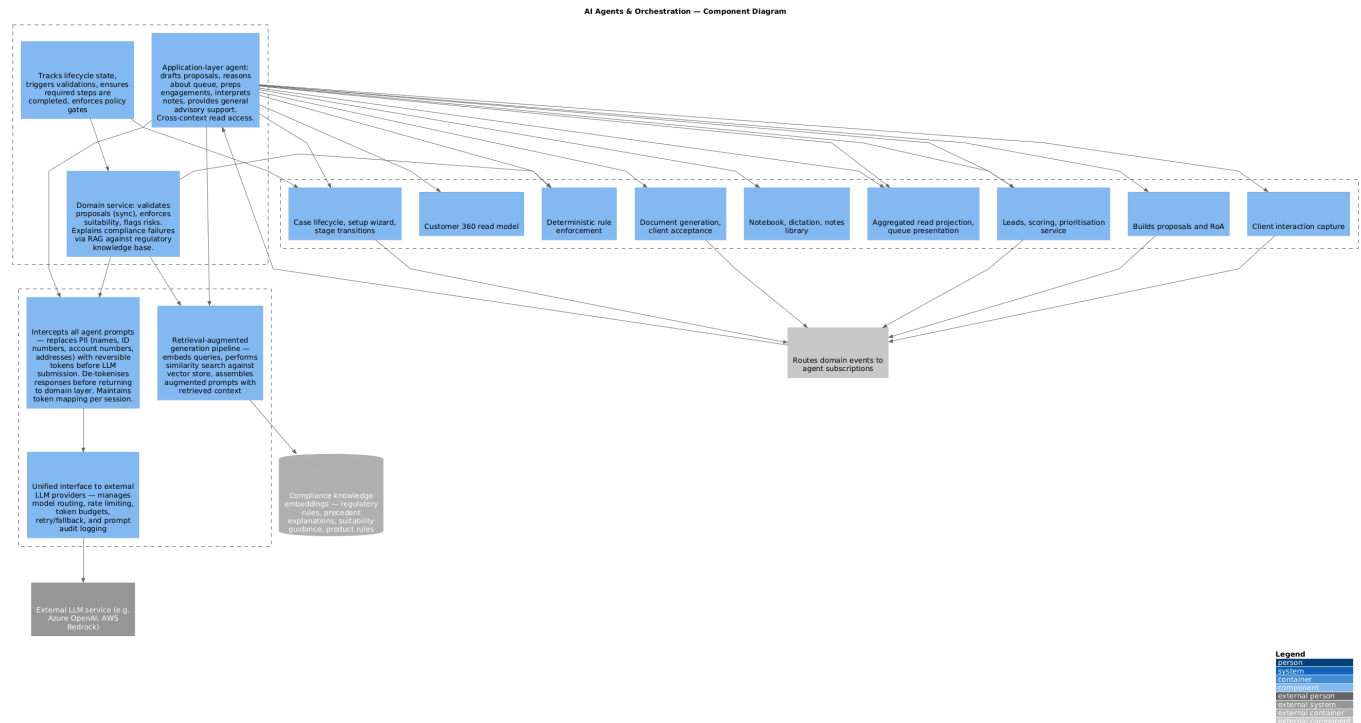
Level 3: Component Diagram – AIR API (Modular Monolith)

Shows the internal module structure of the monolith and its bounded contexts.



Level 3: Component Diagram – AI Agents & Orchestration

Focused view of how Bob and Vera interact with the domain modules. Gary is deferred.

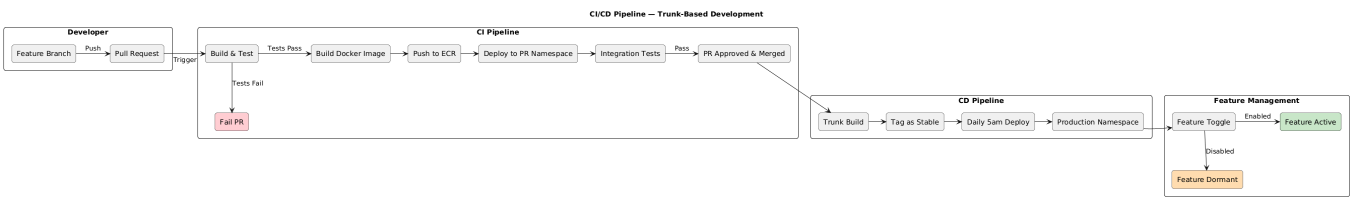


Level 4: Deployment Diagram

Shows the physical deployment topology on AWS. Deployment is infrastructure-level and independent of domain naming.

CI/CD Pipeline Flow

Illustrates the trunk-based development and deployment pipeline. Unchanged from V1.0.



Event Flow Diagram

Shows the event-driven orchestration pattern across bounded contexts.

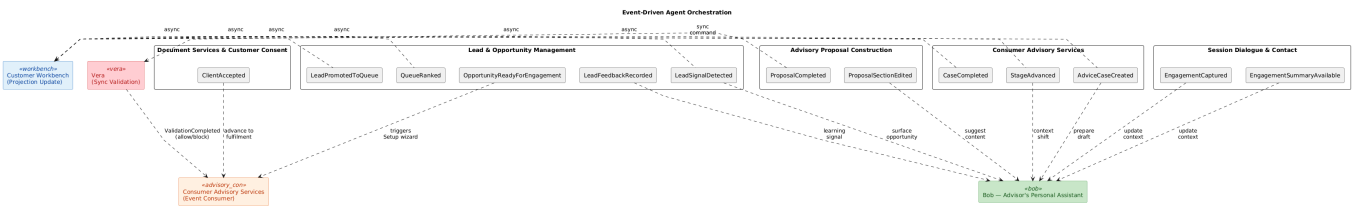


Diagram Summary

Level	Diagram	Purpose
C4 Level 1	System Context	Shows AIR in its ecosystem with users and external systems
C4 Level 2	Container	Shows deployable units – SPA, API monolith, DB, vector store, Bob & Vera agents, PII tokenisation, LLM gateway, event bus
C4 Level 3	Component (API)	Shows internal bounded context modules within the monolith (9 active contexts)
C4 Level 3	Component (Agents)	Shows Bob's cross-context read/write access, Vera's sync validation role, PII tokenisation gateway, RAG pipeline, and LLM integration
C4 Level 4	Deployment	Shows AWS infrastructure topology – EKS, RDS, CloudFront, S3 (uses AWS Architecture Icons)
Supplementary	CI/CD Pipeline	Shows trunk-based development and daily deployment flow
Supplementary	Event Flow	Shows event-driven orchestration with domain event catalogue

Key Architecture Characteristics

- **Modular Monolith:** All bounded contexts deployed as a single unit for speed, with explicit module boundaries for future decomposition
- **9 Active Bounded Contexts:** Customer Workbench, Lead & Opportunity Management, Customer Relationship & Insights, Consumer Advisory Services, Advisory Proposal Construction, Regulatory & Suitability Compliance, Session Dialogue & Contact, Document Services & Customer Consent, Advisor Productivity
- **Naming Convention:** All contexts use canonical service domain names where strong alignment to industry standards exists; bank-specific extensions are clearly documented
- **Bob as Application-Layer Agent:** Cross-context personal assistant with read access to all modules and write access to Advisory Proposal Construction, Customer Workbench, and Lead & Opportunity Management
- **Vera as Domain Service:** Synchronous compliance validation within the Regulatory & Suitability Compliance module; uses RAG against a vector store (pgvector) to explain why compliance checks fail; async monitoring deferred
- **Gary Deferred:** Professional coach agent deferred to future phase (Servicing Activity Analysis context)
- **PII Tokenisation Gateway:** All agent prompts pass through a tokenisation layer that replaces personally identifiable information (names, ID numbers, account numbers, addresses) with reversible session-scoped tokens before submission to LLMs – responses are de-tokenised before returning to the domain layer
- **Vector Store (pgvector):** PostgreSQL with pgvector extension stores compliance knowledge embeddings – regulatory rules, precedent explanations, suitability guidance, and product rules – enabling RAG-based explanations for Vera and contextual support for Bob
- **LLM Gateway:** Unified interface to external LLM providers managing model routing, rate limiting, token budgets, retry/fallback strategies, and prompt audit logging
- **Prioritisation Service in Lead & Opportunity Management:** The Next-Best-Action ranking algorithm lives as a domain service within the Lead & Opportunity module, publishing ranked results to the Customer Workbench projection
- **Opportunity Setup Wizard in Consumer Advisory Services:** Triggered by `OpportunityReadyForEngagement` event from Lead & Opportunity Management, owned by Consumer Advisory Services
- **Event-Driven Backbone:** 13 domain events orchestrate cross-module communication and agent behaviour
- **Policy Gates:** Critical state transitions (e.g. `CompleteProposal`) go through synchronous Vera validation
- **AI Cannot Change State:** Agents suggest, flag, and advise – humans approve and transition
- **Feature Toggles:** Decouple code deployment from business release
- **PR Environments:** Ephemeral namespaces with mock data for rapid feedback
- **Module-Owned Data:** Each module manages its own schema via Liquibase

Glossary

Term	Definition
PEP	Policy Enforcement Point – the enterprise API gateway through which external service calls are routed
AIR	Advice & Intelligence Relationship – the platform being designed
BIAN	Banking Industry Architecture Network – industry-standard reference model for banking service domains
Bob	AI personal assistant agent – advisor's cross-context co-pilot (proposal drafting, queue reasoning, engagement prep, notebook interpretation)
Vera	AI compliance agent – synchronous rule validation, suitability enforcement, and RAG-powered compliance failure explanations
Gary	AI professional coach agent – behavioural analysis and coaching insights (deferred)
RoA	Record of Advice – regulatory documentation of advice given
FNA	Financial Needs Analysis – client assessment inputs
HPA	Horizontal Pod Autoscaler – Kubernetes auto-scaling mechanism
NBA	Next-Best-Action – the prioritised recommendation for what an advisor should do next
QTD	Quarter-to-Date – performance measurement period
GAP	Pre-signature opportunity (not yet in pipeline)
Pipeline	Post-signature, pre-fulfilment opportunity (committed revenue)
CRM	Customer Relationship Management – service domain for managing customer relationships
RAG	Retrieval-Augmented Generation – technique that augments LLM prompts with relevant context retrieved from a vector store to improve accuracy and grounding
pgvector	PostgreSQL extension providing vector similarity search – used as the vector store for agent RAG
PII	Personally Identifiable Information – data that can identify an individual (names, ID numbers, account numbers, addresses)
PII Tokenisation	Process of replacing PII with reversible, session-scoped tokens before data leaves the trust boundary (i.e. before LLM submission)
LLM	Large Language Model – external AI model service used by agents for natural language generation and reasoning

Appendix A: BIAN v14 Service Domain Mapping

The AIR platform's bounded contexts are aligned to the Banking Industry Architecture Network (BIAN) v14 reference model. The table below maps each AIR domain service to its corresponding BIAN service domain(s).

Active Bounded Contexts

AIR Domain Service	Module	BIAN v14 Service Domain(s)	Classification	Notes
Customer Workbench	workbenchModule	Customer Workbench, Point of Service	Core	Owns queue presentation, scorecard, aggregate read projection
Lead & Opportunity Management	leadOpportunityModule	Lead and Opportunity Management, Customer Behavior Insights, Product Matching	Core	Owns lead scoring, pipeline, prioritization service, feedback
Customer Relationship & Insights	customerInsightsModule	CRM, Party Reference Data Directory, Customer Financial Insights	Core (read model)	Read-only projection, upstream client sync
Consumer Advisory Services	advisoryServicesModule	Consumer Advisory Services, Customer Case Management	Core	Owns stage transition, setup wizard, outcome
Advisory Proposal Construction	proposalModule	Consumer Advisory Services (proposal), Suitability Checking, Investment	Core	Owns proposals, RoA, versioning, diffs

AIR Domain Service	Module	BIAN v14 Service Domain(s)	Classification	Notes
		Portfolio Planning		
Regulatory & Suitability Compliance	complianceModule	Regulatory Compliance, Guideline Compliance, Suitability Checking	Core	Sync validation only; async monitoring deferred
Session Dialogue & Contact	sessionDialogueModule	Session Dialogue, Contact Handler, Servicing Event History	Supporting	Structure interaction capture 1 to cases
Document Services & Customer Consent	documentServicesModule	Document Services, Correspondence, Customer Consent, Customer Agreement	Supporting	Document generation, acceptance tracking, compliance artefacts
Advisor Productivity	productivityModule	(No BIAN equivalent – bank-specific extension)	Supporting	Notebook, dictation notes like

Deferred Contexts (not represented in current architecture)

AIR Domain Service	BIAN v14 Service Domain(s)	Reason	Future Home
Party Authentication + Employee Access	Party Authentication, Employee Access	Consumed from bank IAM (Entra ID)	External dependency
Compliance Reporting	Compliance Reporting, Regulatory Reporting	Vera's async mode not yet validated	Future module + event subscriptions
Servicing Activity Analysis (Gary)	Servicing Activity Analysis, Employee Evaluation	Coach agent not in MVP scope	Future module + agent
Deal Execution & Fulfilment	(Absorbed into Consumer Advisory Services)	Terminal lifecycle stage	Extract when complexity warrants

Where AIR Extends Beyond BIAN v14

Extension	AIR Module	Rationale
AI Agent orchestration pattern	Bob, Vera, Gary	BIAN models capabilities, not cross-cutting agent constructs
Next-Best-Action prioritisation	Lead & Opportunity Management (domain service)	BIAN handles lead progression but not ranking algorithms
Personal productivity tools	Advisor Productivity	BIAN excludes internal employee productivity tooling
Event-driven backbone	Cross-cutting	BIAN models capabilities, not the event infrastructure connecting them

Appendix B: Domain Event Catalogue

Complete catalogue of domain events published and consumed across the AIR platform's bounded contexts.

Published Events

Event Name	Publishing Module	Description	Delivery	Key Cons
LeadSignalDetected	Lead & Opportunity Management	Raised when a new behavioural or data signal indicates a potential advisory opportunity for a client	Async	Customer Workbench (project Bob (sur opportur
LeadPromotedToQueue	Lead & Opportunity Management	Raised when a lead passes scoring thresholds and is promoted into the advisor's priority queue	Async	Customer Workbench (project

Event Name	Publishing Module	Description	Delivery	Key Cons
LeadFeedbackRecorded	Lead & Opportunity Management	Raised when an advisor provides feedback (accept, defer, dismiss) on a queued lead	Async	Bob (lead signal)
OpportunityReadyForEngagement	Lead & Opportunity Management	Raised when an opportunity has been fully qualified and is ready for the advisor to begin the engagement process	Async	Consumer Advisory Services (trigger Setup wi
QueueRanked	Lead & Opportunity Management	Raised when the Next-Best-Action Prioritisation Service recalculates queue rankings	Async	Customer Workbench (project
AdviceCaseCreated	Consumer Advisory Services	Raised when a new advice case is instantiated from a qualified opportunity	Async	Customer Workbench (project Bob (pre draft)
StageAdvanced	Consumer Advisory Services	Raised when an advice case transitions to the next lifecycle stage	Async	Customer Workbench (project Bob (cor shift)
CaseCompleted	Consumer Advisory Services	Raised when an advice case reaches its terminal state (fulfilled or withdrawn)	Async	Customer Workbench (project
ProposalSectionEdited	Advisory Proposal Construction	Raised when an advisor (or Bob) edits a section of the advisory proposal	Async	Bob (sug content)

Event Name	Publishing Module	Description	Delivery	Key Cons
ProposalCompleted	Advisory Proposal Construction	Raised when all required proposal sections are complete and the advisor submits for compliance validation	Sync	Regulatory Suitability Compliance (Vera validation gate)
EngagementCaptured	Session Dialogue & Contact	Raised when a client engagement session (call, meeting) is recorded and stored	Async	Bob (update context)
EngagementSummaryAvailable	Session Dialogue & Contact	Raised when a structured summary of a client engagement has been generated (transcript, objectives, outcomes)	Async	Bob (update context)
ClientAccepted	Document Services & Customer Consent	Raised when a client formally accepts the proposed advice (signs RoA or equivalent consent)	Async	Consumer Advisory Services (advance fulfilment Bob (pre draft)
DocumentGenerated	Document Services & Customer Consent	Raised when a final compliance artefact (PDF, RoA document) has been generated	Async	—
ValidationCompleted	Regulatory & Suitability Compliance	Raised when Vera completes a compliance validation pass, carrying an allow or block outcome	Async	Consumer Advisory Services (gate re

Subscriptions by Module

Consuming Module	Subscribed Events	Purpose
Customer Workbench	LeadSignalDetected, LeadPromotedToQueue, QueueRanked, AdviceCaseCreated, StageAdvanced, CaseCompleted	Maintains the aggregated read projection for the advisor's priority queue and active case dashboard
Consumer Advisory Services	OpportunityReadyForEngagement, ValidationCompleted, ClientAccepted	Triggers the Opportunity Setup wizard, processes compliance gate results, advances cases to fulfilment
Regulatory & Suitability Compliance	ProposalCompleted	Triggers synchronous Vera validation as a policy gate before case progression
Bob (AI Agent)	LeadSignalDetected, LeadFeedbackRecorded, AdviceCaseCreated, StageAdvanced, ProposalSectionEdited, EngagementCaptured, EngagementSummaryAvailable, ClientAccepted	Provides contextual awareness for proactive suggestions, draft generation, and engagement preparation
Vera (AI Agent)	ProposalCompleted	Triggers synchronous compliance validation and suitability checking